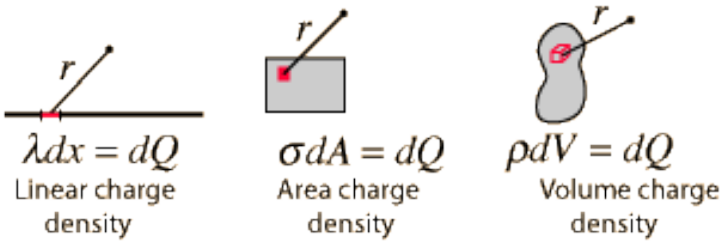
Chapter 36, Electric Fields Part B - Electric field due to a distribution of charge
Calculus Physics II

Charge density (linear λ, surface σ, volume ρ) describes how much charge is packed per unit length, area, or volume, this charge density creates the electric field strength ($\vec{E}$) in the space around it, with the field's strength/direction determined by the charge's distribution. The total electric field at a point in space due to the charge distribution is the vector sum (integral) of the fields from each tiny incremental element dq.



Linear Charge Density (λ – lambda) - is a measure of how electric charge is distributed along a line. It is particularly useful when dealing with charged wires, rods, or any other one-dimensional charge distributions. For a uniform distribution where the total charge (Q) is uniformly distributed along a thin straight rod/wire of length L and is given as

$$\lambda = \frac{Q}{L} \quad units\ are\ \frac{C}{m}$$

And the total charge on the rod/wire would be

$$Q = \lambda L$$

Example: A thin rod of length L = 2 m carries a total charge of 6 μC uniformly distributed.

A) What is its linear charge density?

$$\lambda = \frac{Q}{L} = \frac{6\ x\ 10^{-6} C}{2\ m} = 3\ x\ 10^{-6}\ C/m$$

B) If a wire of length L = 2 m has a linear charge density of 1 x 10^{-6} C/m determine the total charge on the wire.

$$\lambda = \frac{Q}{L} \quad \text{solve for Q}$$

$$Q = \lambda L = (1 \times 10^{-6}\ C/m)(2\ m) = 2 \times 10^{-6} C$$

Non-uniform distribution of surface charge – When dealing with charge distribution that change with the length of the wire the differential formula must be used.

$$\lambda = \frac{dq}{dL}$$

Where the total charge on the conductor is Q, and the incremental charge is dq. The incremental length is dL.

The incremental charge is given by

$$dq = \lambda dL$$



At any point on the length the charge density is given by the ratio of the incremental charge and incremental length. Knowing how the charge changes along the length of the wire, the total charge can be determined by

$$Q = \int \lambda dL$$

Example: A thin straight wire with a length of "L" has a linear charge density that has a dependance along the x-axis of $\lambda = Ax^2$ where A is a constant.

A) What are the units of constant A?

$$\lambda \text{ must have units of } \frac{\text{C}}{\text{m}} \text{ and A is mutiplied by the units } \text{m}^2$$

$$\text{Therefore A must have units of } \frac{\text{C}}{\text{m}^3} \text{ in order to produce the correct units for } \lambda$$

B) Determine the total charge on the wire.

$$Q = \int \lambda dL$$

The length L is along the x-axis therefore dL = dx.

$$Q = \int_0^L \lambda dx = \int_0^L (Ax^2) dx = A \int_0^L x^2 dx$$

$$Q = \frac{A}{3}[x^3]_0^L = \frac{AL^3}{3}$$

Surface/Area Charge Density (σ - sigma) – For a conductor all charge is distributed on the surface of the conductor therefore a conductor with a net charge of Q and a surface area of A, the surface charge density is given as

$$\sigma = \frac{Q}{A} \quad units\ are\ \frac{C}{m^2}$$

For the surface of a conducting sphere the area $A = 4\pi r^2$ therefore

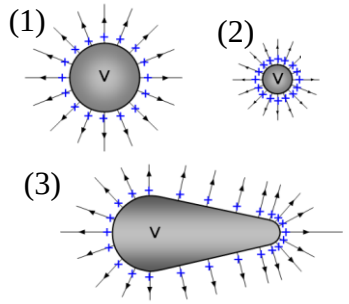
$$\sigma_{Sphere} = \frac{Q}{A} = \frac{Q}{4\pi r^2} \quad - \quad \text{for a conducting sphere}$$

- The larger the radius of the conducting sphere the smaller the surface charge density on that sphere for the same amount of charge.
- For a uniform conducting sphere, the charge density is uniform but if the object has curved surfaces that have different radius, then the surface charge density on the object is not uniform and varies by the surface curvature.

$$\sigma \text{ is proportional to } \frac{1}{r^2}$$

- The smaller the radius the larger the surface charge density.

Example: The diagram below shows three conducting objects with the same net charge but different shapes.



(1) The first sphere has a larger radius than the second sphere (2) therefore the first sphere will have a lower surface charge density and a smaller electric field at the surface.

(2) Smaller radius means larger surface charge density and a larger electric field near the surface.

(3) The larger radius surface to the left will have a lower surface charge density than the smaller radius surface to the right which will have a larger surface charge density therefore the surface charge density of this conducing object is not uniform, and the electric field generated by the net charge will not be uniform, the electric field will be larger at the right and smaller to the left due to the respective charge distributions.

Example: For the conductor's spheres shown in (1) and (2) calculate the charge density on each assuming $r_1 = 10$ cm and $r_2 = 1$ cm and $Q_1 = Q_2 = 10\ \mu C$

$$\sigma_{Sphere\ 1} = \frac{Q}{4\pi r^2} = 7.96x10^{-5} C/m^2 \qquad \sigma_{Sphere\ 2} = \frac{Q}{4\pi r^2} = 7.96x10^{-3} C/m^2$$

- The second sphere has 100 times the surface charge density as the first due to the smaller radius.
- The electric field is created by the surface charge therefore the electric field produced by the charged object will vary with the surface charge!

Non-uniform distribution of surface charge the differential formula must be used.

$$\sigma = \frac{dq}{dA}$$

Where the total charge on the conductor is Q, and the incremental charge is dq. The incremental area is dA.

The incremental charge is given by $dq = \sigma dA$

At any point on the surface the charge density is given by the ratio of the incremental charge and incremental surface area. Knowing how the surface charge is changing the total surface charge can be determined by

$$Q = \iint \sigma dx dy$$

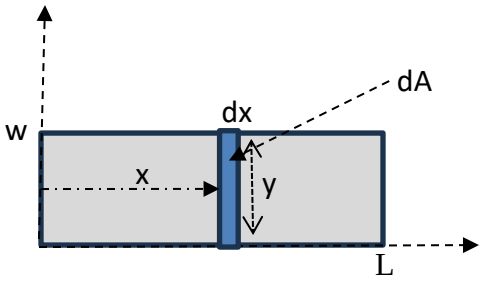
Using symmetries of the object and charge distribution the double integral can become a single integral over the incremental area dA.

$$Q = \int \sigma dA$$

This is an integral across the total area of the conductor.

Example: A rectangular conductor with a length of L and a width of w has a surface charge density that changes along the length as σ = Bx.

Determine the total surface charge on the conductor.



The incremental area dA = length times width = wdx

$$Q = \int \sigma dA = \int_0^L (Bx) w dx = Bw \int_0^L x dx$$

$$Q = \left[\frac{Bwx^2}{2}\right]_0^L = \frac{BwL^2}{2}$$

Volume charge density (ρ - rho) - The volume charge density ρ describes how densely electric charge is distributed within a volume, often used for solid insulators like spheres or cylinders, where Q is the total electric charge, and V is the volume in which the charge is distributed.

$$\text{The volume charge desnity } \rho = \frac{Q_{Total}}{V} \quad units\ of\ \frac{C}{m^3}$$

If the volume charge density changes within the volume and knowing how it changes the total charge can be found by integrating over the volume.

$$Q = \int \rho dV$$

Example: A non-conducting solid sphere of radius R = 10 cm has a uniform volume charge density of ρ = 4 x 10^{-6} C/m^3. Determine the total net charge of the sphere.

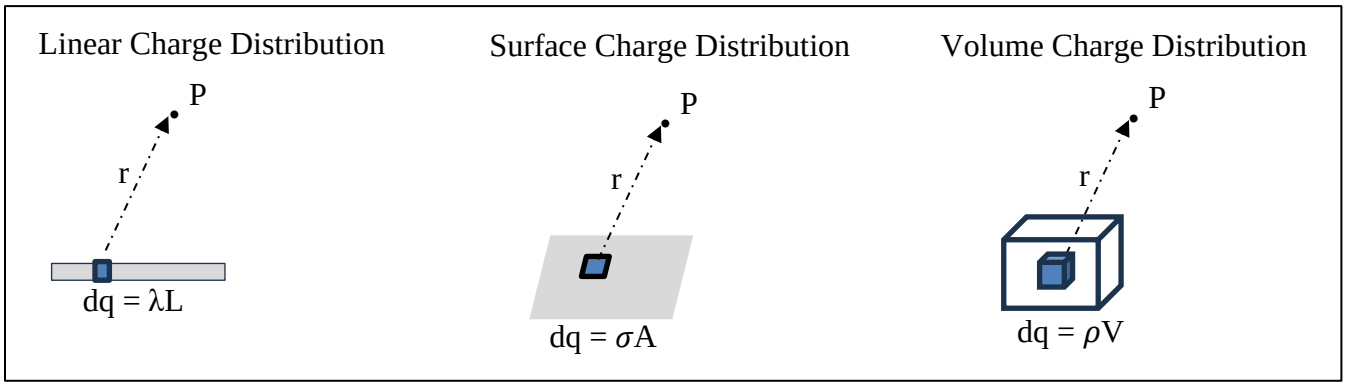
$$\rho = \frac{Q_{Total}}{V} \quad \text{solving for the } Q_{total}$$

$$\text{knowing that a sphere has a volume of } V = \frac{4}{3}\pi R^3$$

$$Q_{Total} = \left(\frac{4}{3}\pi R^3\right)(\rho) = 1.67\ x\ 10^{-8} C$$

Determining the Electric field strength given the charge density - The electric field strength for point charges is the superposition of each field due to a point charge is described by Coulomb's Law. The situation becomes more complex when dealing with continuous charge distributions. These distributions can be linear, surface-based, or volumetric, and require the use of calculus to determine the resulting field. k = 9.0 x 10^9 Nm^2/C^2 and is the electrostatic constant

For an incremental charge dq the electric field at point P from the distribution is shown below.



The electric field due to non-uniform charge distributions:

Linear Charge Density - The linear charge density is λ with units of $\mathrm{C/m}$

The incremental line charge is given as: $dq = \lambda \mathrm{dL}$

Surface Charge Density - The surface charge density is σ with units of $\mathrm{C/m^2}$

The incremental surface charge is given as: $dq = \sigma \mathrm{dA}$

Volume Charge Density - The volume charge density is ρ with units of $\mathrm{C/m^3}$

The incremental volume charge is given as: $dq = \rho \mathrm{dV}$

The electric field $\vec{E}$ at a point in space due to a continuous distribution of charge is found by integrating the contributions from each infinitesimal element of charge dq. The general form of the electric field due to a charge distribution is:

$$Electric\ Field\ General\ Formula \quad \vec{E} = \left(k \int \frac{1}{r^2} dq\right) \hat{r}$$

The Electric Field Strength vectors for various charge distributions.

$$Electric\ Field\ for\ a\ Line\ Charge \ \vec{E} = \left(k \int \frac{\lambda}{r^2} dL\right) \hat{r}$$

$$Electric\ Field\ for\ a\ Surface\ Charge \ \vec{E} = \left(k \int \frac{\sigma}{r^2} dA\right) \hat{r}$$

$$Electric\ Field\ for\ a\ Volume\ Charge \ \vec{E} = \left(k \int \frac{\rho}{r^2} dV\right) \hat{r}$$

Be careful to determine the correct expression for the charge element dq. Also, the radius r is the distance from the incremental charge element dq to the point at which the electric field is being determined. This can get confusing in a circular configuration where the radius of the charge distribution is not the same as the radius from the charge to the point in space where the electric field is being determined.

The electric field is a vector therefore the scalar components of the electric field must be calculated individually then recombined for the total electric field vector at a point in space.

$$\vec{E} = (E_x)\hat{x} + (E_y)\hat{y} + (E_z)\hat{z}$$

Where E_x, E_y and E_z are the scalar components along their respective axis for the electric field vector $\vec{E}$.

$$E_x = \vec{E} \bullet \hat{x} = |\vec{E}| \cos(\alpha) = k \int \frac{1}{r^2} \cos(\alpha)\, dq \;\; along\ the\ dx$$

$$E_y = \vec{E} \bullet \hat{y} = |\vec{E}| \cos(\beta) = k \int \frac{1}{r^2} \cos(\beta)\, dq \;\; along\ the\ dy$$

$$E_z = \vec{E} \bullet \hat{z} = |\vec{E}| \cos(\gamma) = k \int \frac{1}{r^2} \cos(\gamma)\, dq \;\; along\ the\ dz$$

The angle α is measured from the unit vector $\hat{x}$ to the electric field vector
The angle β is measured from the unit vector $\hat{y}$ to the electric field vector
The angle γ is measured from the unit vector $\hat{z}$ to the electric field vector

Steps for solving the electric field strength due to a charge distribution.

1. Define a Coordinate System:

    - Choose a convenient coordinate system (Cartesian, cylindrical, or spherical) that matches the symmetry of the charge distribution. This simplifies calculations.
    - Determine the component of the electric field that is to be calculated and find the projection of the electric field vector along that axis (cos(φ)).

2. Identify the Charge Distribution Type:

    Determine if the charge distribution is:
    - Point Charge: A single, localized charge.
    - Line Charge: A charge distributed along a line.
    - Surface Charge: A charge distributed over a surface.
    - Volume Charge: A charge distributed throughout a volume.

3. Define the relevant charge densities: linear charge density (λ), surface charge density (σ), or volume charge density (ρ).

    - Consider a small incremental charge (dq) within the distribution and sketch on diagram.
    - Draw the radius from the dq to the point in space where the electric field is to be calculated.
    - Express dq in terms of the charge density and the appropriate differential element (dl, dA, or dV).

- Determine which variables change and which are constant over the integration limits.
- Express the differential element in terms of the variables given in the problem.
- Express the variables in terms of the differential element used in the problem using the geometry of the location of the field being determined.

4. Integrate over the Entire Charge Distribution:

   - The total electric field ($\vec{E}$) at the point of interest is found by integrating the electric field contributions from all the small charge elements.
   - Set up the integral using the appropriate limits of integration based on the geometry of the charge distribution.
   - Solve the integral, potentially using symmetrical arguments or coordinate transformations.

5. Apply Superposition (if needed):

   - If the problem involves multiple charge distributions, apply the principle of superposition.

6. Calculate the total electric field due to each distribution separately.

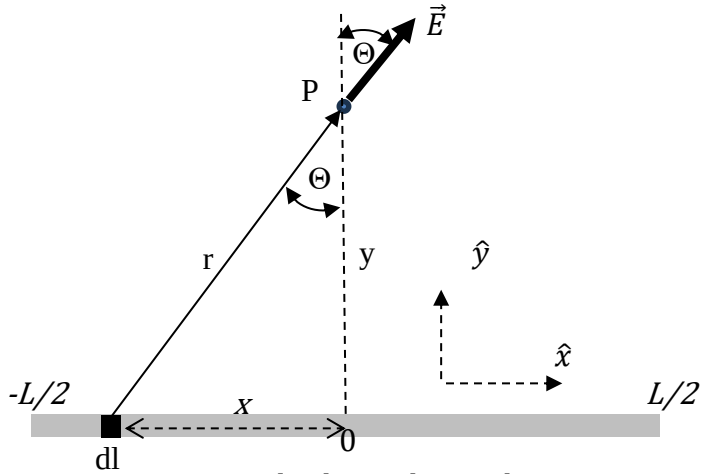
   - Determine the components electric field vector along each of the coordinate system axes to find the total electric field in component form.
   - If necessary, determine the magnitude and direction of the electric field from the components of the electric field vector.

$$\vec{E} = (E_x)\hat{x} + \big(E_y\big)\hat{y} + (E_z)\hat{z}$$

Example: A positively charged thin straight wire has a uniform linear charge density of λ, determine the electric field strength at point P, y meters above the wire and situated at the above the center of the wire of length L. Assume that y and L are constant and do not change in this problem.

$$\vec{E} = (E_x)\hat{x} + \big(E_y\big)\hat{y} + (E_z)\hat{z}$$

Note: The radius r and angle Θ change with the integration along x but the height y above the wire does not.



Positively charged straight wire

Determine the incremental charge on the wire.

$$dq = \lambda dl = \lambda dx$$

Determine the x component of the electric field strength

$$E_x = \vec{E} \bullet \hat{x} = \left|\vec{E}\right| \cos(\alpha) = k \int \frac{1}{r^2} dq \ \ along\ the\ dx$$

$\alpha = 90° - \Theta$ and $\cos(90° - \Theta) = \sin(\Theta)$

$$E_x = \vec{E} \bullet \hat{x} = \left|\vec{E}\right| \sin(\Theta) = k \int \frac{\sin(\Theta)}{r^2} dq$$

Using the right triangle formed by r, x and y

$$r^2 = x^2 + y^2 \quad and \quad \sin(\Theta) = \frac{x}{r} = \frac{x}{(x^2 + y^2)^{1/2}}$$

Substituting in the integral

$$E_x = k \int \frac{\sin(\Theta)}{r^2} (\lambda dx) = \lambda k \int_{-L/2}^{L/2} \left[\frac{x}{(x^2 + y^2)^{1/2}}\right] \left[\frac{1}{(x^2 + y^2)}\right] dx$$

$$E_x = \lambda k \int_{-L/2}^{L/2} \left[\frac{x}{(x^2 + y^2)^{3/2}}\right] dx$$

Solving the integral over the integration limits

$$E_x = \lambda k \left[\frac{1}{(x^2 + y^2)^{1/2}}\right]_{-L/2}^{L/2} = \lambda k \left[\frac{1}{((L/2)^2 + y^2)^{1/2}} - \frac{1}{((-L/2)^2 + y^2)^{1/2}}\right] = 0$$

Determine the y component of the electric field strength.

$$E_y = \vec{E} \bullet \hat{y} = \left|\vec{E}\right| \cos(\beta) = k \int \frac{1}{r^2} dq \ \ along\ the\ dx$$

$\beta = \Theta$ and $\cos(\Theta)$

$$E_y = \vec{E} \bullet \hat{y} = \left|\vec{E}\right| \cos(\Theta) = k \int \frac{\cos(\Theta)}{r^2} dq$$

Using the right triangle formed by r, x and y.

$$r^2 = x^2 + y^2 \quad and \quad \cos(\Theta) = \frac{y}{r} = \frac{y}{(x^2+y^2)^{1/2}}$$

$$E_y = k\int \frac{\cos(\Theta)}{r^2}(\lambda dx) = \lambda k \int_{-L/2}^{L/2} \left[\frac{y}{(x^2+y^2)^{1/2}}\right]\left[\frac{1}{(x^2+y^2)}\right] dx$$

$$E_y = \lambda k y \int_{-L/2}^{L/2} \left[\frac{1}{(x^2+y^2)^{3/2}}\right] dx$$

Solving the integral over the integration limits.

$$E_y = \lambda k y \left[\frac{x}{y^2(x^2+y^2)^{1/2}}\right]_{-L/2}^{L/2} = \left[\frac{\lambda k y}{y^2}\right]\left[\frac{\left(\frac{L}{2}\right)}{\left(\left({}^{L}/_{2}\right)^2 + y^2\right)^{1/2}} - \frac{-\left(\frac{L}{2}\right)}{\left(\left(-{}^{L}/_{2}\right)^2 + y^2\right)^{1/2}}\right]$$

$$E_y = \left[\frac{\lambda k}{y}\right]\left[\frac{\left(\frac{L}{2}\right)}{\left(\left({}^{L}/_{2}\right)^2 + y^2\right)^{1/2}} + \frac{\left(\frac{L}{2}\right)}{\left(\left(-{}^{L}/_{2}\right)^2 + y^2\right)^{1/2}}\right] = \left[\frac{\lambda k}{y}\right]\left[\frac{L}{\left(\left({}^{L}/_{2}\right)^2 + y^2\right)^{1/2}}\right]$$

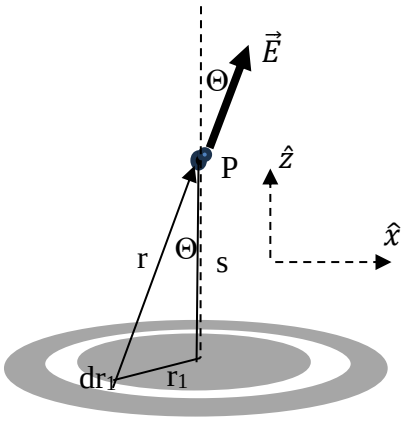
The electric field vector is given as

$$\vec{E} = (0)\hat{x} + \left(\left[\frac{\lambda k}{y}\right]\left[\frac{L}{\left(\left({}^{L}/_{2}\right)^2 + y^2\right)^{1/2}}\right]\right)\hat{y}$$

Example: A flat disk of radius R with a uniform surface charge density of σ is shown below, determine the electric field strength at point P in the $\hat{z}$ direction at a distance of s above the center of the disk.

$$\vec{E} = (E_x)\hat{x} + \left(E_y\right)\hat{y} + (E_z)\hat{z}$$

Note: The radius r and the angle Θ change with the integration limits along r_1 but the height s does not. Also be careful, the radius r from the incremental charge to point P is not the same as r_1, the radius of integration along the disk.



$$E_z = \vec{E} \bullet \hat{z} = \left|\vec{E}\right| \cos(\Theta) = k \int \frac{\cos(\Theta)}{r^2} dq \ \ along\ the\ dz$$

For a surface charge the electric field vector is given by

$$\vec{E} = \left(k \int \frac{\sigma}{r^2} dA\right) \hat{r}$$

Be careful to distinguish between the radius from the charge to the point P, which is r and the distance on the disk to the incremental charge dq which is r_1 and dr_1.

Determine the incremental area dA form dr_1 and the circumstances or the circle shown.

$$dq = \sigma dA = \ \sigma(2\pi r_1) dr_1$$

Using the geometry of the system above to get r and cos(Θ) in terms of r_1.

$$r^2 = {r_1}^2 + s^2 \quad and \ \cos(\Theta) = \frac{s}{r} = \frac{s}{({r_1}^2 + s^2)^{1/2}}$$

Subsituding into the equation

$$E_z = k\sigma 2\pi \int \frac{\cos(\Theta)}{r^2} (r_1 dr_1) = k\sigma 2\pi \int_0^R \left[\frac{s}{({r_1}^2 + s^2)^{1/2}}\right]\left[\frac{1}{({r_1}^2 + s^2)}\right] r_1 dr_1$$

$$E_z = k\sigma 2\pi s \int_0^R \left[\frac{r_1}{({r_1}^2 + s^2)^{3/2}}\right] dr_1 = -k\sigma 2\pi s \left[\frac{1}{({r_1}^2 + s^2)^{1/2}}\right]_0^R$$

$$E_z = -k\sigma 2\pi s \left[\frac{1}{(R^2 + s^2)^{1/2}} - \frac{1}{s}\right]$$

$$\vec{E} = \left(-k\sigma 2\pi s \left[\frac{1}{(R^2 + s^2)^{1/2}} - \frac{1}{s}\right]\right) \hat{z}$$